

Finally, the Era of Unhinged LLM Scaling Comes to an End.

[A Human-Written, Late-Night-Essay by Andreas Gilson]

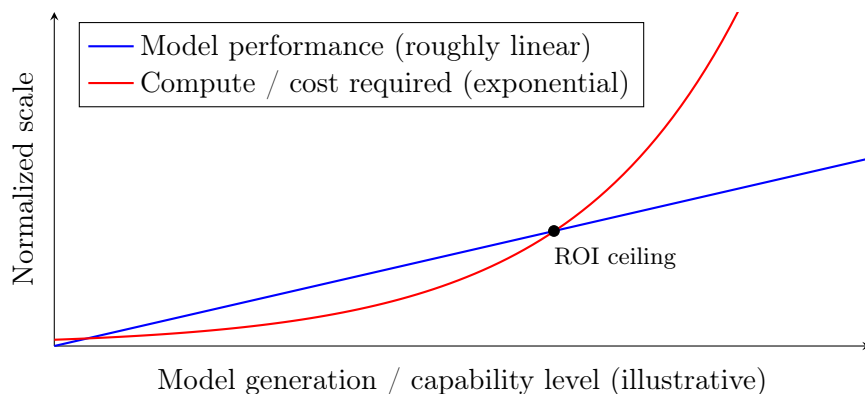
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Sam, Dario, and Elon suddenly agree on slowing things down to save humanity? What is happening? Are we witnessing a paradigm shift where hyper-capitalism finally evolves into an ethically responsible cooperation for the benefit of humankind as a whole? Okay, at least at this point you should realize that this is probably not what's happening. So, what *is* happening?

I don't work on LLM reserach directly, I am not even finished with my PhD in Computer Science but i think about this stuff a lot (and kinda have a degree in philosophy!!), and yet I have the strong urge to share my thoughts with the world about this. A feeling new to me – and believe me, I tried to fight it – but somehow I am strangely confident that sharing these thoughts is the only way to sleep tonight. I am even posting them on (anti)social media, which I despise most of the time.

So here is my take:

Following up on the unfavorable implications of Richard Sutton's "bitter lesson" [1] (more compute will always yield better results), several smart researchers identified multiple scaling laws that formally define this relationship and describe the trade-off between actual progress in LLM research and the cost required to reach the next level of performance, astonishingly well [3, 4]. These are not foundational constants of nature, but they have worked well for describing the development rush of the past years – much like Moore's Law [2], which somehow still seems to hold. In short, scaling laws are the paradigm of the current LLM era, which is precisely the reason why everyone is buying GPUs and building data centers like crazy. However, a simplified reading of these laws is that model performance increases linearly while the required cost and compute increase exponentially. This means that, regardless of the exact variables, there is a mathematical necessity: at some point the return on investment (ROI, for the business folks) of further development will turn negative. In other words, the next marginal improvements of the newer model generation will come at exploding costs, which will never be paid back in revenue.



I ran a few thought experiments,¹ ranging from best guesses of actual current numbers up to a scenario in which every one of the 8.2 billion humans on this planet pays a \$150/month subscription to a single company. The year at which we hit the turning point changes depending on the premises, but the turning point is always there, and surprisingly close: roughly 2028/2029 for the best-guess scenarios, and somewhere in the early-2030s for the extreme “everyone pays” scenario. Because of the linear-versus-exponential relationship between lower model loss (i.e., better model performance) and the cost of getting there, the resulting time frame is surprisingly narrow. Thus, even throwing insane amounts of money at this will only buy you a few extra months.

So what does this mean?

For one, it could explain the odd shift in narrative among hyper-capitalist CEOs, who slowly need to defend their expenses to investors and are starting to worry about having to turn an actual profit on their (still heavily subsidized) product.

It also means that we are close to the LLM ceiling in terms of scaling laws. There is still plenty of room for improvement, but it is no longer unlimited growth. LLMs are already powerful enough to change our society at large, and they will have a huge impact on the world as we know it. But they will also *not* create a super-intelligence that destroys humankind overnight. Even if true intelligence is a purely emergent phenomenon that can arise in any sufficiently complex system or architecture, reaching it will not be economically feasible using autoregressive transformers in the form of LLMs. I know some of you will disagree here, and maybe LLMs will end up being the tool that enables humans to fight each other to extinction some other way. But as long as scaling laws hold, LLMs as we know them will never suddenly spawn an independent entity with power over our species. Sorry to burst your bubble, preppers! Maybe this is at least good news for our planet – imagine the climate impact if the data-center idiocracy never ended.

So what happens to the LLM products and the services built around them? Usually, the next steps in the playbook are clear: after market saturation comes enshittification (hello, ads, and paying to raise the likelihood that an answer mentions your company). But I think it will play out a bit differently this time. This bubble has grown so fast that it is not yet too big to fail (as my wife put it: the German state and its bureaucracy-heavy institutions would not even notice if LLMs disappeared overnight). Enshittifying your product only works once everyone is too dependent on it – or too used to it – to switch. But this time the open-source community (thank you, China, I guess?) is trailing so closely behind the frontier models that many companies and users will simply switch to open- and not-yet-shitty weights that cost a fraction of the frontier models while still solving 90% of the tasks at hand. It might be different for genuinely useful services, or for jobs that AI agents can actually replace in the long run. But in general, this will make it a lot harder to turn subscriptions and expensive tokens into profit, acting as an extended arm of the scaling-law ceiling.

I also believe it won’t take long (or it’s already happening) until many other people reach the same conclusion – especially smart people with capital. So I think the LLM hype is finally slowing down. The bubble will burst, or at least the capital will bleed out slowly. This is not the end of the race toward super-intelligence. The next chapter is probably coming fast (maybe Yann LeCun’s JEPAs will deliver [5]), but for LLMs, this is the beginning of rationalization. Don’t buy the hype. Instead, try to play your part so that the positive aspects of this amazing new technology outweigh the negative ones. This is also part of our responsibility as researchers and as people with deeper insight

¹These are deliberately rough Fermi estimates, as the big players keep their true numbers secret. Of course calculated with an LLM. I see the irony and yet I am not crazy enough to do late night math all by myself.

into the technical side of things – a responsibility I think is currently underrepresented in AI research. But that’s a topic for another essay.

Thank you for reading this, if you made it this far. Please reach out and let’s discuss this together, as humans, using our own brains only.

References

- [1] Richard Sutton. *The Bitter Lesson*. 2019. <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>
- [2] Gordon E. Moore. Cramming more components onto integrated circuits. *Electronics*, 38(8), 1965.
- [3] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, Dario Amodei. *Scaling Laws for Neural Language Models*. <https://arxiv.org/abs/2001.08361>, 2020.
- [4] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, Laurent Sifre. *Training Compute-Optimal Large Language Models*. <https://arxiv.org/abs/2203.15556>, 2022.
- [5] Yann LeCun. *A Path Towards Autonomous Machine Intelligence* <https://api.semanticscholar.org/CorpusID:251881108>, 2022.